

Pre-Calculus Concepts

ANSWER KEY



Limits: an introductory numerical approach

- 1 Estimate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.
- Factor: $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$. As $x \rightarrow 2$, this approaches $2 + 2 = 4$.
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Average rate of change

- 2 Find the average rate of change of $f(x) = x^2 + 1$ over the interval $[1, 4]$.

$$\text{Average rate of change} = \frac{f(4) - f(1)}{4 - 1} = \frac{17 - 2}{3} = \frac{15}{3} = 5.$$

Conic sections: ellipses

- 3 Find the equation of an ellipse centered at the origin with $a = 5$ (horizontal semi-axis) and $b = 3$ (vertical semi-axis).

$$\frac{x^2}{25} + \frac{y^2}{9} = 1.$$

- 4 For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.

Since $16 > 9$, the major axis is horizontal, with $a = \sqrt{16} = 4$. Vertices: $(4, 0)$ and $(-4, 0)$.

Conic sections: hyperbolas

- 5 Find the equation of a hyperbola centered at the origin with vertices at $(\pm 4, 0)$ and $b = 3$.

Since the vertices are horizontal, the equation has the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, with $a = 4$: $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

Parametric equations

- 6 Eliminate the parameter from $x = t + 1$ and $y = t^2$, expressing y in terms of x .

From the first equation, $t = x - 1$. Substitute: $y = (x - 1)^2$.

Sequences and series: sigma notation

- 7 Evaluate $\sum_{k=1}^5 (2k + 1)$.
 $\sum_{k=1}^5 (2k + 1) = 3 + 5 + 7 + 9 + 11 = 35$.
- 8 Write $4 + 8 + 12 + 16 + 20$ using sigma notation, and evaluate it.
This is $\sum_{k=1}^5 4k$. Evaluating: $4(1 + 2 + 3 + 4 + 5) = 4(15) = 60$.
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Binomial theorem

- 9 Use the binomial theorem to expand $(x + 2)^4$.
 $(x + 2)^4 = x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16 = x^4 + 8x^3 + 24x^2 + 32x + 16$.
- 10 Find the coefficient of x^3 in the expansion of $(x + 3)^5$, using $\binom{5}{k}$.
The term with x^3 comes from $\binom{5}{3}x^3(3)^2 = 10x^3(9) = 90x^3$. The coefficient is 90.
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Vectors in the plane

- 11 Given $\mathbf{u} = (3, -4)$, find its magnitude and a unit vector in the same direction.
 $|\mathbf{u}| = \sqrt{9 + 16} = 5$. Unit vector: $\frac{1}{5}(3, -4) = (0.6, -0.8)$.
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A multi-part conic and function analysis problem

- 12 This question has two parts. A circle has equation $x^2 + y^2 - 4x + 6y - 12 = 0$. (a) Complete the square to write the equation in standard form, and state the center and radius. (b) Determine whether the point $(6, -1)$ lies inside, on, or outside the circle.
- (a) Group and complete the square: $(x^2 - 4x + 4) + (y^2 + 6y + 9) = 12 + 4 + 9$, giving $(x - 2)^2 + (y + 3)^2 = 25$. Center: $(2, -3)$. Radius: 5. (b) Distance from center to $(6, -1)$: $\sqrt{(6 - 2)^2 + (-1 - (-3))^2} = \sqrt{16 + 4} = \sqrt{20} \approx 4.47$. Since $4.47 < 5$, the point lies INSIDE the circle.
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Matrix operations

- 13 Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find $A - B$.
- Subtract corresponding entries: first row $(2 - 0, -1 - 5) = (2, -6)$; second row $(3 - (-2), 4 - 1) = (5, 3)$.
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Infinite geometric series

- 14 Find the sum of the infinite geometric series $8 + 4 + 2 + 1 + \dots$, using $S = \frac{a_1}{1-r}$.
First term $a_1 = 8$, ratio $r = \frac{1}{2}$. Sum $= \frac{8}{1 - \frac{1}{2}} = \frac{8}{\frac{1}{2}} = 16$.
- 15 Write the repeating decimal $0.\overline{6}$ as a fraction, by treating it as an infinite geometric series $\frac{6}{10} + \frac{6}{100} + \frac{6}{1000} + \dots$.
First term $a_1 = \frac{6}{10}$, ratio $r = \frac{1}{10}$. Sum $= \frac{6/10}{1 - 1/10} = \frac{6/10}{9/10} = \frac{6}{9} = \frac{2}{3}$.
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Piecewise and step functions in applied contexts

- 16 A shipping company charges 5 dollars per pound for the first 10 pounds, then 3 dollars per pound for each pound beyond 10. Write a piecewise function $C(w)$ for the cost, and find the cost of a 15-pound package.
 $C(w) = 5w$ for $0 < w \leq 10$, and $C(w) = 50 + 3(w - 10)$ for $w > 10$. For 15 pounds:
 $C(15) = 50 + 3(5) = 50 + 15 = 65$ dollars.
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Finding terms of a sequence defined recursively

- 17 A sequence is defined by $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$. Find a_4 .
 $a_2 = 2(3) + 1 = 7$. $a_3 = 2(7) + 1 = 15$. $a_4 = 2(15) + 1 = 31$.
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A multi-part conic section identification and analysis problem

- 18 This question has three parts. An equation is given: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. (a) Complete the square in both x and y to write the equation in standard conic form. (b) Identify the type of conic section. (c) State the center and the lengths of the semi-axes.
(a) Group terms: $4(x^2 - 4x) + 9(y^2 + 2y) = 11$. Complete the square:
 $\frac{4(x^2 - 4x + 4)}{(x-2)^2} + \frac{9(y^2 + 2y + 1)}{(y+1)^2} = 11 + 16 + 9$, giving $4(x-2)^2 + 9(y+1)^2 = 36$. Divide by 36:
 $\frac{4}{36} + \frac{9}{36} = 1$. (b) Since both terms are positive and added (not subtracted) with different denominators, this is an ellipse. (c) Center: $(2, -1)$. Semi-axes: $a = \sqrt{9} = 3$ (horizontal), $b = \sqrt{4} = 2$ (vertical).
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Multiple choice — easy

19 Find the average rate of change of $f(x) = 2x^2 - 3$ over the interval $[-3, 2]$.

a. -10

b. 0

c. -2 ✓

d. -7

$$f(2) = 5 \text{ and } f(-3) = 15. \text{ Average rate of change} = \frac{f(2) - f(-3)}{2 - (-3)} = \frac{5 - 15}{5} = -2.$$

20 Find the average rate of change of $f(x) = 3x^2 - 7x + 5$ over the interval $[0, 3]$.

a. 5

b. 2 ✓

c. 6

d. -1

$$f(3) = 11 \text{ and } f(0) = 5. \text{ Average rate of change} = \frac{f(3) - f(0)}{3 - 0} = \frac{11 - 5}{3} = 2.$$

21 Find the average rate of change of $f(x) = 2x^2 - 7x + 9$ over the interval $[1, 3]$.

a. 3

b. -1

c. 2

d. 1 ✓

$$f(3) = 6 \text{ and } f(1) = 4. \text{ Average rate of change} = \frac{f(3) - f(1)}{3 - 1} = \frac{6 - 4}{2} = 1.$$

22 Find the average rate of change of $f(x) = 3x^2 - 7x - 9$ over the interval $[0, 4]$.

a. 8

b. 1

c. 5 ✓

d. 20

$$f(4) = 11 \text{ and } f(0) = -9. \text{ Average rate of change} = \frac{f(4) - f(0)}{4 - 0} = \frac{11 + 9}{4} = 5.$$

23 Find the average rate of change of $f(x) = 3x^2 + 8x - 9$ over the interval $[5, 6]$.

a. 41 ✓

b. 43

c. 40

d. 44

$$f(6) = 147 \text{ and } f(5) = 106. \text{ Average rate of change} = \frac{f(6) - f(5)}{6 - 5} = \frac{147 - 106}{1} = 41.$$

24 Find the average rate of change of $f(x) = 3x^2 - 5x - 1$ over the interval $[-2, -1]$.

a. -15

b. -12

c. -11

d. -14 ✓

$$f(-1) = 7 \text{ and } f(-2) = 21. \text{ Average rate of change} = \frac{f(-1) - f(-2)}{-1 + 2} = \frac{7 - 21}{1} = -14.$$

25 Find the average rate of change of $f(x) = x^2 + 7x$ over the interval $[-6, 3]$.

a. 4 ✓

b. 5

c. 36

d. -5

$$f(3) = 30 \text{ and } f(-6) = -6. \text{ Average rate of change} = \frac{f(3) - f(-6)}{3 + 6} = \frac{30 + 6}{9} = 4.$$

26 Find the average rate of change of $f(x) = -x^2 - 3x - 3$ over the interval $[-4, 2]$.

- a. -7
- b. -1 ✓
- c. -2
- d. -6

$$f(2) = -13 \text{ and } f(-4) = -7. \text{ Average rate of change} = \frac{f(2) - f(-4)}{2 - (-4)} = \frac{-13 - (-7)}{6} = -1.$$

27 Find the average rate of change of $f(x) = x^2 - 9x + 7$ over the interval $[-3, 4]$.

- a. -8 ✓
- b. -56
- c. -15
- d. -7

$$f(4) = -13 \text{ and } f(-3) = 43. \text{ Average rate of change} = \frac{f(4) - f(-3)}{4 - (-3)} = \frac{-13 - 43}{7} = -8.$$

28 Find the average rate of change of $f(x) = -3x^2 + 7x - 2$ over the interval $[1, 4]$.

- a. -11
- b. -6
- c. -24
- d. -8 ✓

$$f(4) = -22 \text{ and } f(1) = 2. \text{ Average rate of change} = \frac{f(4) - f(1)}{4 - 1} = \frac{-22 - 2}{3} = -8.$$

29 Find the average rate of change of $f(x) = -3x^2 + 2x + 9$ over the interval $[0, 1]$.

- a. -4
- b. -1 ✓
- c. -2
- d. 1

$$f(1) = 8 \text{ and } f(0) = 9. \text{ Average rate of change} = \frac{f(1) - f(0)}{1 - 0} = \frac{8 - 9}{1} = -1.$$

30 Find the average rate of change of $f(x) = -x^2 + 6x - 3$ over the interval $[1, 6]$.

a. -1 ✓

b. -2

c. -5

d. -6

$$f(6) = -3 \text{ and } f(1) = 2. \text{ Average rate of change} = \frac{f(6) - f(1)}{6 - 1} = \frac{-3 - 2}{5} = -1.$$

31 Find the average rate of change of $f(x) = x^2 - 7$ over the interval $[-2, 1]$.

a. 0

b. -3

c. -4

d. -1 ✓

$$f(1) = -6 \text{ and } f(-2) = -3. \text{ Average rate of change} = \frac{f(1) - f(-2)}{1 - (-2)} = \frac{-6 - (-3)}{3} = -1.$$

32 Find the average rate of change of $f(x) = -2x^2 - 6x + 7$ over the interval $[5, 6]$.

a. -28 ✓

b. -29

c. -30

d. -26

$$f(6) = -101 \text{ and } f(5) = -73. \text{ Average rate of change} = \frac{f(6) - f(5)}{6 - 5} = \frac{-101 - (-73)}{1} = -28.$$

33 Find the average rate of change of $f(x) = x^2 + 2x + 7$ over the interval $[3, 4]$.

a. 11

b. 10

c. 8

d. 9 ✓

$$f(4) = 31 \text{ and } f(3) = 22. \text{ Average rate of change} = \frac{f(4) - f(3)}{4 - 3} = \frac{31 - 22}{1} = 9.$$

Multiple choice — medium

34 Evaluate $\sum_{k=1}^{10} (k - 1)$.

a. 46

b. 44

c. 45 ✓

d. 55

$$\sum_{k=1}^{10} (k - 1) = \sum_{k=1}^{10} k - 10 = 1(55) - 10 = 45.$$

35 Evaluate $\sum_{k=1}^4 (2k + 4)$.

a. 40

b. 34

c. 20

d. 36 ✓

$$\sum_{k=1}^4 (2k + 4) = 2 \sum_{k=1}^4 k + 16 = 2(10) + 16 = 36.$$

36 Evaluate $\sum_{k=1}^{10} (2k - 2)$.

a. 90 ✓

b. 88

c. 100

d. 110

$$\sum_{k=1}^{10} (2k - 2) = 2 \sum_{k=1}^{10} k - 20 = 2(55) - 20 = 90.$$

37 Evaluate $\sum_{k=1}^4 (k - 4)$.

a. -7

b. -2

c. 10

d. -6 ✓

$$\sum_{k=1}^4 (k - 4) = \sum_{k=1}^4 k - 16 = 1(10) - 16 = -6.$$

38 Evaluate $\sum_{k=1}^{10} (k + 3)$.

a. 85 ✓

b. 84

c. 95

d. 55

$$\sum_{k=1}^{10} (k + 3) = \sum_{k=1}^{10} k + 30 = 1(55) + 30 = 85.$$

39 Evaluate $\sum_{k=1}^{11} (k - 2)$.

a. 55

b. 66

c. 44 ✓

d. 43

$$\sum_{k=1}^{11} (k - 2) = \sum_{k=1}^{11} k - 22 = 1(66) - 22 = 44.$$

40 Evaluate $\sum_{k=1}^8 (k - 5)$.

a. -5

b. 4

c. -4 ✓

d. 36

$$\sum_{k=1}^8 (k - 5) = \sum_{k=1}^8 k - 40 = 1(36) - 40 = -4.$$

41 Evaluate $\sum_{k=1}^8 (k - 3)$.

a. 11

b. 12 ✓

c. 36

d. 20

$$\sum_{k=1}^8 (k - 3) = \sum_{k=1}^8 k - 24 = 1(36) - 24 = 12.$$

42 Evaluate $\sum_{k=1}^{11} (2k + 1)$.

a. 132

b. 154

c. 143 ✓

d. 141

$$\sum_{k=1}^{11} (2k + 1) = 2 \sum_{k=1}^{11} k + 11 = 2(66) + 11 = 143.$$

43 Evaluate $\sum_{k=1}^6 (-3k - 2)$.

a. -69

b. -72

c. -75 ✓

d. -63

$$\sum_{k=1}^6 (-3k - 2) = -3 \sum_{k=1}^6 k - 12 = -3(21) - 12 = -75.$$

44 Evaluate $\sum_{k=1}^{11} (-3k + 2)$.

a. -176 ✓

b. -173

c. -165

d. -198

$$\sum_{k=1}^{11} (-3k + 2) = -3 \sum_{k=1}^{11} k + 22 = -3(66) + 22 = -176.$$

45 Evaluate $\sum_{k=1}^8 (-2k + 5)$.

a. -24

b. -30

c. -72

d. -32 ✓

$$\sum_{k=1}^8 (-2k + 5) = -2 \sum_{k=1}^8 k + 40 = -2(36) + 40 = -32.$$

46 Evaluate $\sum_{k=1}^{10} (3k - 3)$.

a. 135 ✓

b. 145

c. 165

d. 132

$$\sum_{k=1}^{10} (3k - 3) = 3 \sum_{k=1}^{10} k - 30 = 3(55) - 30 = 135.$$

47 Evaluate $\sum_{k=1}^{12} (3k + 5)$.

a. 306

b. 291

c. 294 ✓

d. 234

$$\sum_{k=1}^{12} (3k + 5) = 3 \sum_{k=1}^{12} k + 60 = 3(78) + 60 = 294.$$

48 Evaluate $\sum_{k=1}^5 (-3k + 2)$.

a. -32

b. -45

c. -35 ✓

d. -30

$$\sum_{k=1}^5 (-3k + 2) = -3 \sum_{k=1}^5 k + 10 = -3(15) + 10 = -35.$$

Multiple choice — hard

49 For the ellipse $\frac{x^2}{1681} + \frac{y^2}{81} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 9, 0)$

b. $(0, \pm 40)$

c. $(\pm 40, 0)$ ✓

d. $(\pm 41, 0)$

Since $a^2 = 1681$ and $b^2 = 81$: $c^2 = a^2 - b^2 = 1681 - 81 = 1600$, so $c = \sqrt{1600} = 40$. Foci: $(\pm 40, 0)$

- 50 For the ellipse $\frac{x^2}{841} + \frac{y^2}{400} = 1$ (horizontal major axis), find the coordinates of the foci.
- a. $(\pm 29, 0)$
 - b. $(0, \pm 21)$
 - c. $(\pm 21, 0)$ ✓
 - d. $(\pm 20, 0)$

Since $a^2 = 841$ and $b^2 = 400$: $c^2 = a^2 - b^2 = 841 - 400 = 441$, so $c = \sqrt{441} = 21$. Foci: $(\pm 21, 0)$.

- 51 For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ (horizontal major axis), find the coordinates of the foci.
- a. $(\pm 4, 0)$ ✓
 - b. $(\pm 3, 0)$
 - c. $(0, \pm 4)$
 - d. $(\pm 5, 0)$

Since $a^2 = 25$ and $b^2 = 9$: $c^2 = a^2 - b^2 = 25 - 9 = 16$, so $c = \sqrt{16} = 4$. Foci: $(\pm 4, 0)$.

- 52 For the ellipse $\frac{x^2}{1369} + \frac{y^2}{1225} = 1$ (horizontal major axis), find the coordinates of the foci.
- a. $(\pm 37, 0)$
 - b. $(0, \pm 12)$
 - c. $(\pm 35, 0)$
 - d. $(\pm 12, 0)$ ✓

Since $a^2 = 1369$ and $b^2 = 1225$: $c^2 = a^2 - b^2 = 1369 - 1225 = 144$, so $c = \sqrt{144} = 12$. Foci: $(\pm 12, 0)$.

- 53 For the ellipse $\frac{x^2}{1369} + \frac{y^2}{144} = 1$ (horizontal major axis), find the coordinates of the foci.
- a. $(\pm 12, 0)$
 - b. $(\pm 37, 0)$
 - c. $(0, \pm 35)$
 - d. $(\pm 35, 0)$ ✓

Since $a^2 = 1369$ and $b^2 = 144$: $c^2 = a^2 - b^2 = 1369 - 144 = 1225$, so $c = \sqrt{1225} = 35$. Foci: $(\pm 35, 0)$.

- 54 A hyperbola has vertices at $(\pm 20, 0)$ and $b = 21$. Find the coordinates of the foci.
- a. $(\pm 29, 0)$ ✓
 - b. $(\pm 20, 0)$
 - c. $(\pm 21, 0)$
 - d. $(\pm 41, 0)$

$c = \sqrt{a^2 + b^2} = \sqrt{400 + 441} = \sqrt{841} = 29$. Foci: $(\pm 29, 0)$.

- 55 A hyperbola has vertices at $(\pm 3, 0)$ and $b = 4$. Find the coordinates of the foci.
- a. $(\pm 5, 0)$ ✓
 - b. $(\pm 7, 0)$
 - c. $(\pm 3, 0)$
 - d. $(\pm 4, 0)$

$c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Foci: $(\pm 5, 0)$.

56 A hyperbola has vertices at $(\pm 9, 0)$ and $b = 40$. Find the coordinates of the foci.

- a. $(\pm 9, 0)$
- b. $(\pm 41, 0)$ ✓
- c. $(\pm 40, 0)$
- d. $(\pm 49, 0)$

$$c = \sqrt{a^2 + b^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41. \text{ Foci: } (\pm 41, 0).$$

57 A hyperbola has vertices at $(\pm 12, 0)$ and $b = 35$. Find the coordinates of the foci.

- a. $(\pm 12, 0)$
- b. $(\pm 35, 0)$
- c. $(\pm 37, 0)$ ✓
- d. $(\pm 47, 0)$

$$c = \sqrt{a^2 + b^2} = \sqrt{144 + 1225} = \sqrt{1369} = 37. \text{ Foci: } (\pm 37, 0).$$

58 A hyperbola has vertices at $(\pm 8, 0)$ and $b = 15$. Find the coordinates of the foci.

- a. $(\pm 15, 0)$
- b. $(\pm 8, 0)$
- c. $(\pm 23, 0)$
- d. $(\pm 17, 0)$ ✓

$$c = \sqrt{a^2 + b^2} = \sqrt{64 + 225} = \sqrt{289} = 17. \text{ Foci: } (\pm 17, 0).$$

- 59 Find the sum of the infinite geometric series with first term $a_1 = 12$ and common ratio $r = \frac{1}{3}$, using $S = \frac{a_1}{1-r}$.

a. 9

b. 36

c. 18 ✓

d. 8

$$S = \frac{a_1}{1-r} = \frac{12}{2/3} = 18.$$

- 60 Find the sum of the infinite geometric series with first term $a_1 = 11$ and common ratio $r = -\frac{1}{4}$, using $S = \frac{a_1}{1-r}$.

a. $\frac{55}{4}$

b. $\frac{44}{5}$ ✓

c. -44

d. $\frac{44}{3}$

$$S = \frac{a_1}{1-r} = \frac{11}{5/4} = \frac{44}{5}.$$

- 61 Find the sum of the infinite geometric series with first term $a_1 = 20$ and common ratio $r = \frac{3}{4}$, using $S = \frac{a_1}{1-r}$.

a. 80 ✓

b. $\frac{80}{7}$

c. 5

d. $\frac{80}{3}$

$$S = \frac{a_1}{1-r} = \frac{20}{1/4} = 80.$$

- 62 Find the sum of the infinite geometric series with first term $a_1 = 12$ and common ratio $r = \frac{1}{4}$, using $S = \frac{a_1}{1-r}$.

a. 9

b. 48

c. $\frac{48}{5}$

d. 16 ✓

$$S = \frac{a_1}{1-r} = \frac{12}{3/4} = 16.$$

- 63 Find the sum of the infinite geometric series with first term $a_1 = 15$ and common ratio $r = \frac{3}{4}$, using $S = \frac{a_1}{1-r}$.

a. $\frac{15}{4}$

b. 20

c. 60 ✓

d. $\frac{60}{7}$

$$S = \frac{a_1}{1-r} = \frac{15}{1/4} = 60.$$